

Task 1 – RTL Design & Functional Simulation (Verilog)

Candidate Information

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Internship: Very Large Scale Integration (VLSI)

Platform: InternSpark

Goal

Design and verify an 8-bit ALU using Verilog HDL and perform functional simulation using testbench and waveform analysis.

Requirements

- Create synthesizable Verilog/SystemVerilog modules
- Write comprehensive testbench
- Simulate using EDA Playground / EPWave
- Verify waveform outputs

RTL Design Description

An 8-bit ALU was designed using synthesizable Verilog HDL. The ALU supports arithmetic, logical, shift, and comparison operations based on a 3-bit opcode input.

Supported Operations

Opcode	Operation
000	Addition
001	Subtraction
010	AND
011	OR
100	XOR
101	Left Shift

Opcode	Operation
110	Right Shift
111	Compare

RTL Source File

Design File: design.sv

RTL Code

```
// Code your design here
module alu_8bit(
    input [7:0] A,
    input [7:0] B,
    input [2:0] opcode,
    output reg [7:0] result,
    output reg carry,
    output reg zero
);
always @(*) begin
    carry = 0;
    case(opcode)
        3'b000: begin
            {carry, result} = A + B;
        end
        3'b001: begin
            {carry, result} = A - B;
        end
        3'b010: begin
            result = A & B;
        end
        3'b011: begin
            result = A | B;
        end
        3'b100: begin
            result = A ^ B;
        end
        3'b101: begin
            result = A << 1;
        end
        3'b110: begin
            result = A >> 1;
        end
        3'b111: begin
            result = (A == B) ? 8'b00000001 : 8'b00000000;
        end
        default: begin
            result = 8'b00000000;
        end
    endcase
    zero = (result == 8'b00000000);
end
endmodule
```

Figure 1: RTL Design

Testbench Design

A SystemVerilog testbench was created to verify all ALU operations. Different combinations of inputs and opcodes were applied to validate arithmetic and logical functionality.

Testbench File: testbench.sv

Testbench Code

```
`timescale 1ns/1ps
module tb_alu_8bit;
reg [7:0] A;
reg [7:0] B;
reg [2:0] opcode;
wire [7:0] result;
wire carry;
wire zero;
alu_8bit uut (
    .A(A),
    .B(B),
    .opcode(opcode),
    .result(result),
    .carry(carry),
    .zero(zero)
);
initial begin
    $dumpfile("alu.vcd");
    $dumpvars(0, tb_alu_8bit);
    // ADD
    A = 8'd10; B = 8'd5; opcode = 3'b000;
    #10;
    // SUB
    A = 8'd20; B = 8'd4; opcode = 3'b001;
    #10;
    // AND
    A = 8'b11001100; B = 8'b10101010; opcode = 3'b010;
    #10;
    // OR
    opcode = 3'b011;
    #10;
    // XOR
    opcode = 3'b100;
    #10;
    // LEFT SHIFT
    A = 8'd8; opcode = 3'b101;
    #10;
    // RIGHT SHIFT
    opcode = 3'b110;
    #10;
    // COMPARE
    A = 8'd15; B = 8'd15; opcode = 3'b111;
    #10;
    $finish;
end
endmodule
```

Figure 2: Testbench

Simulation Procedure

The RTL design and testbench were compiled and simulated using Icarus Verilog on EDA Playground.

Simulation Commands

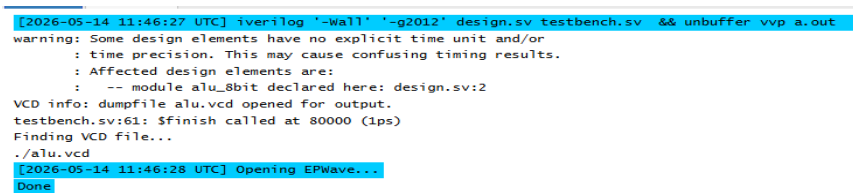
```
iverilog -o alu_sim design.sv testbench.sv
```

```
vvp alu_sim
```

Simulation Output

The simulation completed successfully and generated waveform data for functional verification.

Simulation Console Screenshot



```
[2026-05-14 11:46:27 UTC] iverilog '-wall' '-g2012' design.sv testbench.sv && unbuffer vvp a.out
warning: Some design elements have no explicit time unit and/or
: time precision. This may cause confusing timing results.
: Affected design elements are:
:   -- module alu_8bit declared here: design.sv:2
VCD info: dumpfile alu.vcd opened for output.
testbench.sv:61: $finish called at 80000 (1ps)
Finding VCD file...
./alu.vcd
[2026-05-14 11:46:28 UTC] Opening EPWave...
Done
```

Figure 3: Simulation Output

Waveform Analysis

Waveforms were analyzed using EPWave to verify ALU functionality. The testbench module (tb_alu_8bit) and Unit Under Test (uut) were expanded to observe signal transitions and output behavior.

Signals verified:

- A
- B
- opcode
- result
- carry
- zero

Waveform Output

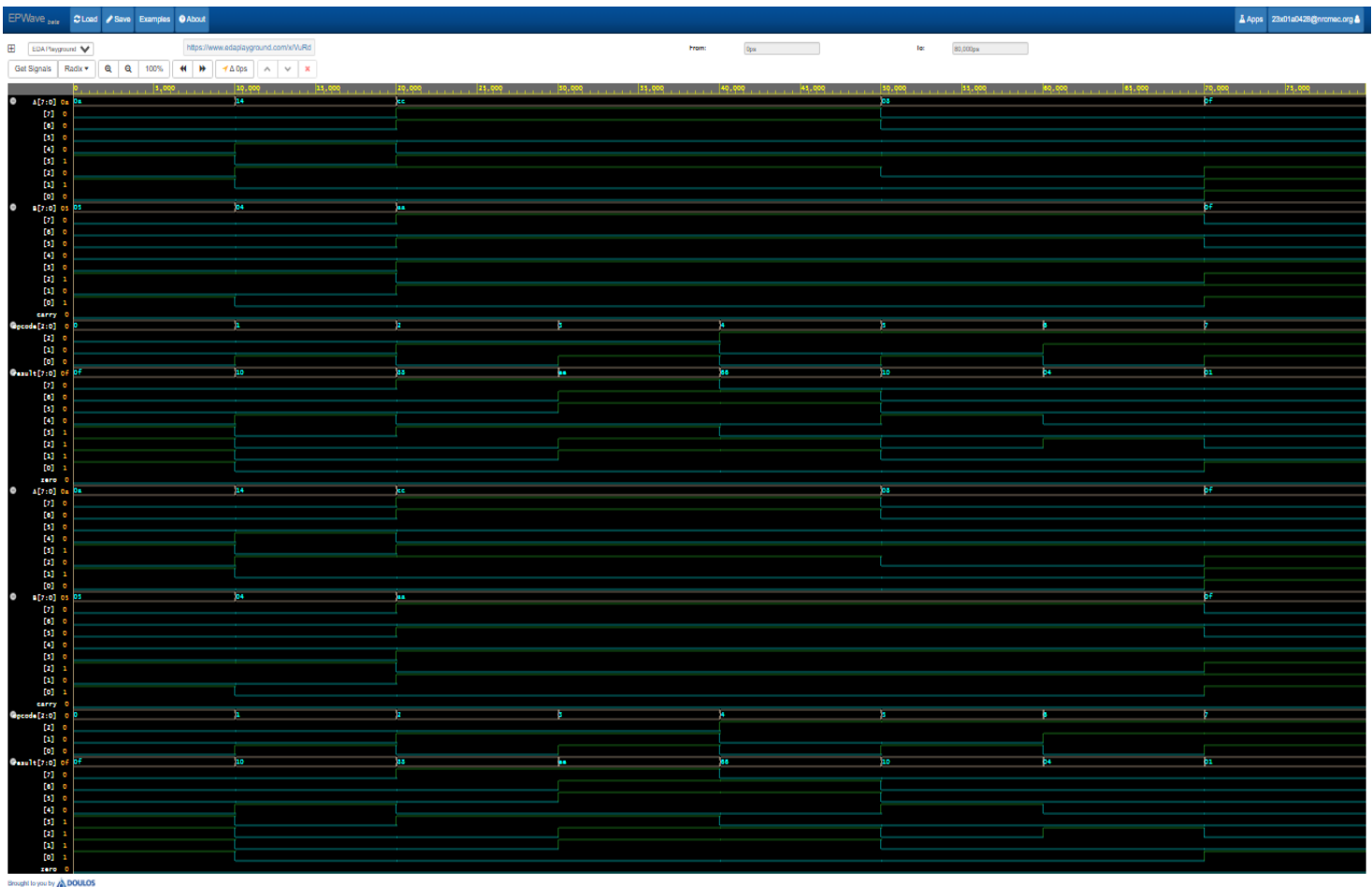


Figure 4: Waveform Output

Functional Verification Results

The waveform confirms correct ALU functionality for all arithmetic and logical operations.

Opcode	Operation	Output Verification
000	Addition	Verified
001	Subtraction	Verified
010	AND	Verified
011	OR	Verified
100	XOR	Verified
101	Left Shift	Verified
110	Right Shift	Verified
111	Compare	Verified

Live Demo:

<https://www.edaplayground.com/x/VuRd>

Conclusion

The 8-bit ALU was successfully implemented using Verilog HDL and verified through functional simulation. Arithmetic, logical, shift, and comparison operations were validated using waveform analysis. The simulation results confirmed correct functionality for all supported ALU operations.

GitHub Task Completion Link

https://github.com/PavanCh26/InternSpark_VLSI/tree/main/Task1